TRACE-RPG: Source-Bound Ontology Evidence and a Bounded Codex-Assisted Integration Case in an Interactive Game

Anonymous Author(s)

***Abstract*—TRACE-RPG combines external authorization, typed-error repair, unchanged fallback, revalidation, replay, and accounting; its read-only bilingual ontology view binds source declarations to public observations without mutation authority. One operator-assisted Codex dashboard case (n=1, independent repeats 0) reports `gpt-6.1-sol`, repaired geometry, retained failures, 62 checks, a Godot Web build, and real-input denial/disclosure/redisplay; provider identity, revision, seed, cost, request count, and inference latency remain unavailable. A fixed-game native pilot on one authored quest records 8 assigned rows (7 observed): 5 system-goal successes, one retained pre-response harness-serializer technical failure (executed `null`, not rerun), and 2 `gpt-6.1-sol` ontology response-timeout failures; 23 host invocations, 0 repairs, 0/16 unsafe committed transitions, 7 valid semantic-link replays, and 6 valid fresh-engine reexecutions (1 `NOT_RUN`; first row unobserved). It is exploratory, with one stateless intent proposer and deterministic planner/checker; it does not establish A/C, ontology efficacy, ranking, noninferiority, or player benefit.**

***Index Terms*—game AI, interactive narrative, symbolic validation, constrained generation, reproducibility**

I. Introduction

GENERATED events may request absent items, unknown facts, forbidden disclosures, unauthorized effects, or regressions. Plausibility [1], [2] and grounding [3], [4] do not authorize mutation: **only TRACE-RPG's external deterministic gate authorizes mutation** (Fig. 1).

Table I scopes **C1—contracts**, **C2—validation/repair**, **C3—audit links**, **C4—assignment accounting**, and **C5—guided repair** to their evidence and inference ceilings.

II. Related Work and Research Gap

Grounding/memory/planning/role [3], [4], [5], [6] and player/dialogue work [7], [8], [9], [10] distinguish believability from admissibility; affect/engagement [11], [12], [13] remains future, unimplemented, and non-authoritative.

Benchmarks span state, embodiment, content, process, social interaction, and long-horizon play [1], [2], [14], [15], [16], [17], [18], [19], [20]; NCP-Bench tests narrative commitment/fact consistency, not player outcomes [21]. Structural/schema, affordance, and rule methods [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33] motivate constraints, not superiority; retrieval/memory supplies context without commit authority [34], [35], [36].

Manuscript prepared for anonymous review.

TABLE I
Contributions, Prior Topics, Evidence Lanes, and Inference Ceilings (Generated from the Structured Matrices)

ID	Contribution	Prior topics (refs)	Lanes	Inference ceiling
C1	Trust-boundary contracts	T1, T2, T3 (17)	E1, ENG1	encoded-field contract conformance
C2	Validate–repair–commit controller	T1, T8 (14)	E1, ENG1	mechanism behavior on frozen cases
C3	Audit-linked evidence layer	T4, T7, T8 (6)	E1, ENG1	integrity/replay on named fixtures
C4	Assignment-complete harness	T1, T4, T6, T7 (20)	E1	exact designed-case accounting
C5	Counterexample-guided operator ρ	T1, T8, T9 (8)	E1	frozen offline repairability classes

T1 = grounded game worlds; T2 = retrieval/memory agents; T3 = structured generation; T4 = environments/benchmarks; T5 = player experience; T6 = human/LLM evaluation; T7 = statistics/reproducibility; T8 = planning/runtime interposition; T9 = feedback repair. Lanes are defined in Table V. All rows are authored-fixture verified; none supports an efficacy claim.

Reproducibility/evaluation motivates frozen artifacts, retained failures, construct/rater/bias controls, and justified margins/random effects [37], [38], [39], [40], [41], [42], [43], [44], [45]. Replay/leakage checks support denominators, not efficacy [46], [47]. C1–C5 combine enforcement, state/transaction management, feedback repair, action interposition, preconditions/effects, abort, social/belief-aware planning, and replay [27], [28], [29], [48], [46], [49], [50], [51], [52], [53], [54]; no invention/firstness claim.

III. Problem Definition and Guarantee Boundary

$c_t = (G_t, q_t)$ stores typed snapshot G_t and external action-policy/disclosure/quest-stage constraints. Predicates neither read $h_{<t}$ nor reconstruct `WorldState`; transitions update G_t only, excluding policy rotation. Retrieval, scores, rationales, memory, and future affect lack authority.

Strict parsing of untrusted candidate a_t enforces the shared key/schema contract, rejecting unknown top-level fields. Validation returns

$$\begin{aligned} V(c_t, a_t) &= (\boldsymbol{v}, E_t), \\ \boldsymbol{v} &= (v_{\text{policy}}, v_{\text{pre}}, v_{\text{reach}}, v_{\text{know}}, \\ &\qquad v_{\text{disc}}, v_{\text{elig}}, v_{\text{stage}}) \in \{0,1\}^7, \end{aligned} \tag{1}$$

where E_t is a structured error set. Mutation is

$$c_{t+1} = \begin{cases} T(c_t, a_t), & \bigwedge_{i=1}^{7}(v_i = 1), \\ c_t, & \text{otherwise.} \end{cases} \tag{2}$$

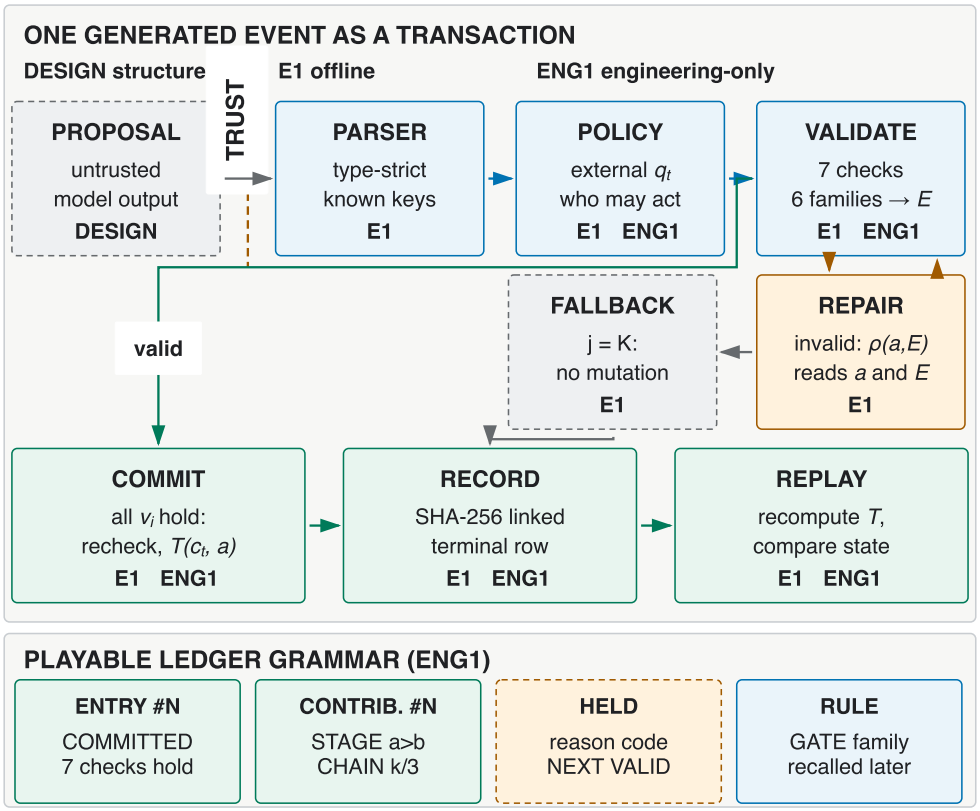


Fig. 1. Policy/c_t gates proposals; errors feed bounded ρ or unchanged fallback. Passing candidates are rechecked, committed, recorded, and replayed. Historical cohorts/legacy ENG1 remain archived, not active model evidence. Below: committed-snapshot ledger grammar.

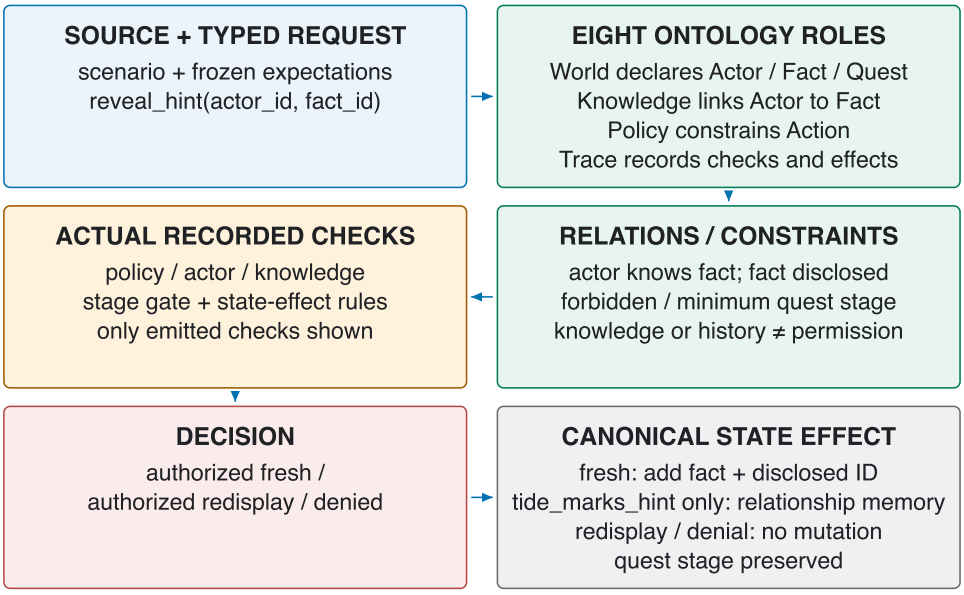


Fig. 2. Eight-role ontology structure (Table III) links source fields to checks and observed decisions. Only authorized transitions change canonical state; the diagram is neither live capture nor hidden-state evidence.

TABLE II
ENCODED VALIDATOR BOUNDARY

Predicate	Rejects	On failure
Action policy	unknown action/type; omitted required preconditions/effects; disallowed effects or quest-stage effects	unchanged
Precondition	absent/false requirement	unchanged
Reachability	inaccessible object	unchanged
NPC knowledge	undeclared known fact	unchanged
Disclosure	forbidden fact	unchanged
Quest stage (`quest_stage`)	required stage above current	unchanged
Quest-stage monotonicity (`quest_stage_monotonicity`)	regressive stage	unchanged

Fixtures test invariants, not theorems. **I1:** validation exhaustion or pre-commit proposal/controller failure preserves state. **I2:** returning callbacks and $K \geq 0$ bound attempts to $K+1$ with pre-commit revalidation; without deadlines, wall-clock termination is not guaranteed. **I3:** accepted `CandidateAction.effects` stay within `ActionPolicy.allowed_effects`, including authorized quest-stage targets. **I4:** frozen schemas/deterministic transitions enable terminal-state reconstruction.

Assumptions: correct policy/extraction, deterministic versions, returning callbacks, single-process ownership. Replay checks structured transitions, not repair provenance, prompts, stochastic generation, or undeclared natural-language implications.

IV. TRACE-RPG ARCHITECTURE AND ALGORITHMS

A. Trust Boundary and Contracts

External models/adapters propose; parser rejects ambiguous types/unknown keys; independent policy defines preconditions/effects. Versioned bridge carries observations, proposals, validations, rejections, fallbacks, and commits (Fig. 1); unknown-key parity proves neither parser equivalence nor portability. Python has exactly seven deterministic checks: `action_policy`, `precondition`, `object_reachability`, `npc_knowledge`, `forbidden_disclosure`, `quest_stage`, and `quest_stage_monotonicity`. Godot's six state-relative families and seven DISC rules are distinct, non-isomorphic labels; its `QUEST STAGE` display spans two Python quest checks.

The role map `game-track/web/dashboard/ontology-map.json` binds declarations/frozen expectations; below, I abbreviates `initial_state`. Table III's `reveal_hint(actor_id,fact_id)` separates view, repair, and authority: the view exposes `tide_marks_hint`/stage-2 policy; ρ sees only a_t, E_t, may insert a named predicate, and cannot read `WorldState`. Godot commits only all-seven-pass candidates; fresh disclosure increments revision $+1$; redisplay does not.

TABLE III
EIGHT ONTOLOGY ROLES: READ-ONLY SOURCE BINDINGS, NOT LIVE INFERENCE OR AUTHORITY

Role	Source field or relationship example
World	I.`world.object_locations`: `signal_lens`→`lamp_store`.
Actor	I.`npcs.captain_mira`; sole `scope.eligible_actor_ids` member.
Knowledge	Actor `knowledge`→canonical Fact IDs, not permission.
Fact	I.`facts`: `dock_saved`, `lighthouse_dark`; `keeper_betrayal` is an ID, not payload.
Policy	`disclosure_policy`: forbids `keeper_betrayal`; `tide_marks_hint` needs minimum quest stage 2.
Quest	I.`quest`: stage 0, flag `dock_saved`; disclosure preserves stage.
Action	`reveal_hint(actor_id,fact_id)` adds Fact/disclosed ID; only tide hint adds relationship memory. Fresh revision $+1$; redisplay unchanged.
Trace	`prior_state_hash`, `next_state_hash`, `rule_ids`, `template_id`, `case_id` bind packet/version to evidence; absent fields cannot establish PASS.

$a \leftarrow \text{Propose}(c)$; $j \leftarrow 0$
repeat
  $(\text{ok}, E) \leftarrow V(c, a)$; $\text{Record}(a, E)$
  if ok **then**
    assert $V(c, a).\text{ok}$; **return** $\text{Commit}(c, a)$
  if $j = K$ **then return** $\text{Fallback}(c)$
  $a \leftarrow \text{Repair}(c, a, E)$; $j \leftarrow j + 1$
until terminal

Fig. 3. Validate–repair–commit. At most $K+1$ candidate attempts are recorded; success defensively validates before mutation.

B. Fixed-goal NPC integration

Under `npc-goal-recovery/1.1.0`, Codex proposes intent, Python checks public claims/plans, and only Godot writes state. The fixed harbor-restoration/authorized-tide-lead goal is not model-editable. Bindings cover session, epoch, request, scenario, NPC, goal, version, policy/config hashes, revision/state hash, deadline/budget, and issued action. Service: `127.0.0.1:8768`; browser origins: `localhost`, `127.0.0.1:8877`.

1) NPC-DISC-1 disclosure contract and authority boundary: NPC-DISC-1's 60 declaration-derived `reveal_hint` cases do not rerun/enlarge historical `verified-run-006`. Generated results/Tables X–XII define strata, bindings, and input-to-check mappings; Godot rechecks service bindings before mutation.

C. Validate–Repair–Commit

In Fig. 3's dependency-free pseudocode, Record appends in memory; persistence follows terminal consistency. (ok, E) abbreviates V with $\text{ok} := \bigwedge_{i=1}^{7}(v_i = 1)$; $V(c, a).\text{ok}$ revalidates.

Repair cannot mutate state; exhaustion preserves it. Typed `counterexample_guided_repair` reads only candidate a/errors E (its sole state-derived input), never `WorldState`:

$$\rho(a, E) = a \oplus \bigcup_{e \in E} \delta(e), \tag{3}$$

TABLE IV
PER-ERROR-CLASS EDITS OF THE GUIDED OPERATOR ρ

Validator error class	**Edit $\delta(e)$ on the candidate**
Policy precondition omission	insert the named predicate
Policy effect omission	insert the named effect
Policy effect violation	drop the named effect
Unsatisfied precondition	drop the named predicate
Unauthorized stage mutation	clear `quest_stage_effect`
Unknown action type	no-op (no registered alternative)
Unreachable object; knowledge, disclosure, and stage classes	no-op (candidate edit would launder or needs state)

TABLE V
EVIDENCE LANES AT A GLANCE: DESIGN, UNIT, COMPARISON, AND CEILING (GENERATED FROM THE EXPERIMENT MATRIX)

Lane	**Design**	**Unit / budget**	**Comparison**	**Ceiling**
E1	Offline conformance: authored frozen fixtures, one world	13 gate; 12 repair/arm; 10 faults; 7 adapter; 3 guards	reject vs blind vs ρ vs oracle	mechanism conformance
E3	KG/ontology simulation: closed-world typed-link holdout	43 nodes; 106 ref. edges; 24 typed; 210 scores	degree baseline vs 6 fixed strategies	construction result only
ENG1	Engineering-only Godot/Web: engine-local policy mirror, no participants	4 fixtures; 52 checks; 8 smoke; 5 archetypes	authored fixture paths	engineering-only conformance

Active lanes are non-interchangeable with different denominators; the historical E2 receipts are excluded from active generation and ENG1 is engineering-only. No active lane establishes population efficacy, model ranking, player experience, retrieval benefit, or commercial-engine performance.

Table IV's $\oplus$ applies deterministic $\delta(e)$: one field/error entity, at most $|E|$/attempt; conflicting requirements/rejections leave fields unchanged. Fixed points require identical E, not convergence. The nondeployable state-reading oracle ignores errors, reconstructs preconditions/effects and both quest-stage fields from state, and never launders violations by deleting required objects/used facts/disclosed facts.

D. Integrity, Semantic Replay, and Accounting

Unkeyed SHA-256 detects accidental/tested changes, not identity or joint rewriting. Replay checks prior-state/candidate/validation, invalid nonfinal attempts, recomputed final state, and JSONL continuity, never repair. Request→decision→result and first-action/one-receipt bindings do not prevent duplicate IDs or impersonation. Terminal classes are commit, symbolic fallback, adapter failure, and controller failure (state/assignment preserved; no completed proposal trace). Pre-execution run/arm/scenario/seed/model/revision keys plus configuration/input/prior-state hashes bind assignments; aggregation rejects duplicate/missing/unexpected keys and separates failures.

V. EVIDENCE LANES AND BOUNDARIES

A. *E1* Offline Conformance Design

PQ1–PQ5 cover immutability, repair, checksum/semantic/linkage/continuity faults, rollback, fail-closed assignment/adapter/manifest accounting, and unknown-key rejection. Results enumerate 13 gate fixtures (1 valid, 12 negative) and 12 repair fixtures; latency/token constants test accounting, not population/detector performance. Packets bind fixtures, assignments, schemas, results, and code.

TABLE VI
SIMULATION-ONLY PROPOSAL-STRATEGY EVALUATION ON A FROZEN TYPED-LINK HOLDOUT

Strategy	Precision	Recall	MRR	Brier	Sem@3
S0 degree baseline	0.000	0.000	0.282	0.315	0.556
S2 typed-lexical	1.000	1.000	0.944	0.131	1.000

B. Archived live-screening materials

Archived live-screening is immutable provenance/hash evidence, excluded from active tables/counts/claims and never transferred to `gpt-6.1-sol`.

C. NPC-ENG1/NPC-SCR1: 88 Assignments, 28 Definitions

Stable v5 has 16 engineering assignments and 12 screening recipes ($12 \times 2 \times 3 = 72$; combined $R/P/O = 40/24/24$), covering bootstrap, identity/hash, intent/plan/role/result, negative/stale/duplicate/deadline, and replay. P uses authenticated Codex OAuth without fallback; O is 100% auxiliary-source, not independent gold. Status/count/latency/reproduction: `research/academic-pipeline/npc-goal-recovery/verified-run-006/`; verified-run-002/003/004 are historical. ENG-X01/SCR-X01, D02/QA-001, C-RESULT, Frozen8/Frozen16, v1–v4, and `native012` are provenance, not C6 evidence.

D. *E3* KG/Ontology Link-Proposal Simulation

E3 is an engineering-only, non-authoritative closed-world simulation: six authored graph-edge-withheld targets select/summarize among seven fixed strategies, not an untouched or external test. **Scope.** This table reports authored closed-world link recovery, not runtime KG retrieval, semantic completeness, long-horizon contradiction reduction, or evidence for any efficacy claim. The deterministic methods graph contains 43 graph nodes, 106 reference edges, and 24 curated typed edges; all ontology checks pass. The generated table is construction evidence, not OWL/SHACL conformance, runtime retrieval, semantic completeness, or efficacy.

VI. PILOT RESULTS

A. Declared-Field Gate Conformance

The loader admits only one valid control plus one isolated instance of each implemented validator code; therefore 13/13 agreement and all 12 implemented codes are *harness construction invariants*, not measured success. Beyond this setup, 12 negative fixtures matched authored expected codes rather than another code, and every noncommit path preserved complete canonical state. This is single-world-state structured-field-oracle conformance, not independent semantic-label accuracy.

TABLE VII
E1 OFFLINE CONFORMANCE: EXACT REPAIR-ARM COUNTS BY REPAIRABILITY CLASS (DESIGNED FIXTURES)

Arm	Info source	Commits / cases			Mean edits[e]
		G-rep.[f]	O-only[f]	Irrep.[f]	Fields
Rejection-only (K=0)	none	0/5	0/1	0/6	—
Blind unchanged retry	none	0/5	0/1	0/6	—
Validator-error-guided ρ	error set E	5/5	0/1	0/6	1.0
State-reading oracle	state+policy	5/5	1/1	0/6	1.0

[e] Mean changed candidate fields over committed repairs (minimality proxy); failed arm executions left the state unchanged. E1 uses 12 initially invalid designed cases: rejection-only/blind 0/12, 0/12; validator-error-guided ρ 5/12; state-reading oracle 6/12 commits. [f] Commits/cases. G-rep. = guided-repairable, O-only = oracle-only, Irrep. = irreparable (frozen repairability classes); these are not population efficacy or live-model quality.

TABLE VIII
E1 OFFLINE CONFORMANCE: FROZEN-PILOT CHECKS AND TERMINAL OUTCOMES (ROWS ARE NOT COMPARABLE)

Check or terminal family	Exact outcome	Interpretation
Gate fixture agreement	13/13	authored-oracle agreement
Known-boundary acceptance	2/2	encoded limit, not a safety pass
Closed unknown-key rejection	1/1	proposal/replay parser parity
Detectable integrity-fault rejection	10/10	designated check operations
Known provenance-boundary acceptance	1/1	repair provenance unauthenticated
Adapter terminal classes	1 commit; 1 fallback; 5 failures / 7 assigned	complete classification of 7 assignments
Manifest-guard rejection	3/3	injected faults failed closed

Rows have different denominators and favorable directions; they must not be ranked as rates. Every value is an exact audit fact for designed fixtures.

B. Repair-Arm Comparison on Designed Fixtures

Frozen classes for 12 initially invalid designed cases are 5 guided-repairable, 1 oracle-only, and 6 irreparable. Rejection-only and blind unchanged-candidate retry each committed 0/12. ρ committed 5/5, 0/1, and 0/6, respectively, editing a mean of 1.0 fields per commit. It reads only the prior candidate and typed validator errors E, its sole state-derived input; failed arms preserve canonical state. The state-reading reference callback committed 5/5, 1/1, and 0/6, respectively. Blind never converts errors, ρ handles only payload-sufficient cases, and the oracle adds state knowledge. The 5 committed guided cases initially carried `POLICY_PRECONDITION_OMISSION+UNSATISFIED_PRECONDITION`, `POLICY_EFFECT_OMISSION`, `POLICY_EFFECT_VIOLATION`, `UNSATISFIED_PRECONDITION`, `POLICY_QUEST_STAGE_EFFECT_VIOLATION`, respectively, and each became valid through the corresponding Table IV edit $\delta(e)$ alone. Frozen-fixture operator conformance, not live-model repair quality.

C. Integrity, Boundaries, and Assignment Accounting

All 10 prespecified detectable faults were rejected by designated checks (10/10); without stable typed detector codes, no detector-layer attribution follows. A known provenance-boundary fixture changed the invalid precursor before the same recorded valid repair, recomputed the checksum, and passed replay (1/1 undetected): replay neither authenticates repair generation nor blocks writers recomputing unkeyed checksums. Two open sentinels—unextracted narrative disclosure and simultaneous candidate/policy omission of a required object—were accepted at the encoded layer (2/2), with 0 safety passes. The former unknown-top-level acceptance became a closed regression: proposal and replay parsers rejected a complete 12-field candidate with one unknown key (1/1), showing parser rejection parity, not semantic safety. Seven adapter/accounting assignments yielded 1 commit, 1 symbolic fallback, and 5 adapter failures; synthetic telemetry was populated and propagated for 2/7 (schema-pinned accounting, not measurement). All 3/3 injected duplicate/missing-assignment guards failed closed; these are raw counts from synthetic offline fixtures.

D. Fixed-Goal NPC Operation (NPC-ENG1/NPC-SCR1)

Mira delivers the authorized tide lead after harbor-signal restoration. Python computes bounded BFS plans from typed intent/public observations; players acquire/install, while the NPC commits only the authorized `reveal_hint`. Godot synchronously checks session/epoch/request/goal/policy/revision/hash before calling the existing writer without an await; results/new observations choose the next plan, not single-candidate validator-error repair.

NPC-ENG1 has 16 deterministic R-arm engineering cases from actual engine snapshots (assigned_failure=2, completed=14); NPC-SCR1 retains all 72 P/R/O assignments and failures, 24 per arm (assigned_failure=12, completed=60). P records 24 authenticated Codex CLI exec attempts, 24 returned decisions, and 4 assigned failures—not remote billing-call counts. Requested model: unknown (null); recorded decision-model identifier: unknown (null). The latter is the configured selector, not a provider revision; provider revision is unavailable. Null identifiers remain unknown; global defaults are not evidence. R uses authored rules. O is a protocol-declared auxiliary under `same_model_family_auxiliary`. Actual model-family matching is unverified. No source is independent; O is neither an oracle nor semantic gold. Table IX gives bounded protocol-conformance fixed-case counts, not recovery-efficacy, superiority, or generalization estimates; efficacy pending, tokens/cost unmeasured. A read-only CLI and isolated working directory were used; tool-free inference and adversarial confidentiality were not established. Immutable verified-run-006 is revalidated against archived source bytes dated 2026-10-17 (revision fa9fb8a); this historical packet result does not certify the current checkout NPC runtime or current release authority. Verified pre-authored stale-control ACK rejections are P=4, R=4, O=4 per arm (SCR-T01 (stale_revision), SCR-T02 (stale_policy)); 12 instances had all corresponding negative-control gates passed.

TABLE IX
NPC-SCR1 FIXED-CASE EXECUTION COUNTS, SEPARATE FROM THE 16 NPC-ENG1 CASES; ALL ASSIGNED FAILURES REMAIN IN THE DENOMINATOR.

Arm	Assigned	Complete/fail	Goal reached	Recovery E/U
P	24	20/4	4	24/0
R	24	20/4	4	24/0
O	24	20/4	4	24/0

E: recovery eligible; U: unreachable; NA: unmeasured. Observed goal counts are not recovery rates. Valid replay does not turn a failed assignment into a completed one. O has zero independent sources.

Assigned failures remain in the denominator, not method-performance outcomes; other assigned failures (0) remain unclassified. Replay validation is limited to the exported epoch (`intent_lineage_scope=epoch_local_only`); it does not prove intent continuity across epochs (`cross_epoch_intent_continuity_proven=false`). The opaque-anchor exception allows only an adjacent leading observation request/decision pair in a positive-epoch snapshot; epoch-zero orphans, other observation requests before intent commitment, and later intent mismatches remain rejected.

E. Frozen `reveal_hint` contract and disclosure evidence

Only the frozen expectation packet `sl-npc-disclosure-expectations-v1` (v1.0.0) and four result files are consumed. Packet SHA-256 `9920aa834910a3ee35ed4446de4fdba42bb67dda71b07a93bccfc7a0b2e09028` was checked against its sidecar. The declared normal stratum has 48 cases under `1 eligible actor x 4 declared fact IDs x 3 declared stage values x 2 knowledge values x 2 prior-disclosed values`; boundary 4 and malformed 8 are excluded. Trace expectations match packet case IDs, not gate output.

Content-first explanation maps typed input (operation, actor, fact, stage, knowledge, prior disclosure) through ontology entities/relations and authored rules to checks, an allowed/denied decision, and canonical state effect. A permitted first display adds only the requested fact (plus the declared relationship-memory entry for the tide hint), preserves stage, and increments revision once; denial/redisplay preserves canonical state. Explanatory only: no hidden facts, natural-language implications, or live capture.

Runtime declaration: `sl-npc-disclosure-expectations-v1:runtime-disclosure` / v1.0.0, provenance `frozen_expectation_packet`. Scope contains only four canonical fact IDs (dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint) and eligible actor `captain_mira`. Rule IDs: DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007; approved templates only `mira-safe-deflection-v1` and `sl-rc-003-authorized-hint`.

Python evaluator materializes the same cases/expectation object in three isolated arms, recording semantic outcomes, check traces, errors, before/after state SHA-256, rendered template IDs, and replay equality. Godot separately owns the existing `reveal_hint` writer and canonical mutation at the game-execution boundary; this producer neither reconstructs nor replaces Godot state. Traceability binds operation `reveal_hint`, scenario `sealed-lighthouse-v1`, actor `captain_mira`, and bound source hashes. Execution-observed hashes match the current source. 3 source hash(es) differ from the frozen baseline (game-track/godot/scripts/game3d/game_3d.gd, game-track/godot/scripts/sealed_lighthouse_machine.gd, llm-wiki/wiki/reports/2026-10-22-ontology-signals-interview.md); the packet hash is preserved and the difference is disclosed.

TABLE X
FROZEN REVEAL_HINT CONTRACT STRUCTURE, CONTENT, RULE IDS, AND RESULT PROVENANCE.

Field	Value
Operation	reveal_hint
Scenario	sealed-lighthouse-v1
Normal denominator	48
Boundary	4
Malformed	8
Canonical facts	dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint
Eligible actor	captain_mira
Ontology	sl-npc-disclosure-expectations-v1:runtime-disclosure
Rule IDs	DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007
Templates	mira-safe-deflection-v1; sl-rc-003-authorized-hint

Expectations are not derived from gate output; denominators are read from the frozen declaration.

Table XI reports the three arms over the same 60 cases with separate normal/boundary/malformed denominators; unchecked violations are intentional isolated observations. The checked general-rule and ontology-gate arms have identical outcomes on all 60 of 60 authored cases. The rule-change fixture set contains no discriminating case between these arms: all 2 fixtures and their 4 paired cases match. This bounds authorization conformance under the shared frozen rules, not general ontology equivalence or ontology superiority. Figure 4 distinguishes E/R (expected-match/replay cases), C (check-record PASS/FAIL), and NR (cases with no reported checks); NR is not skipped, pass, or out-of-scope. Table XII maps rule-change fixtures and paired cases to disposable policy clones; the shipped scenario is unchanged.

Claims are limited to bounded engineering conformance: no human gold labels, access-separated holdout, efficacy, superiority, generalization, player benefit, model latent state, or cost. Exact input paths, hashes, and reproduction commands are bound by `paper/latex/generated/disclosure_producer_manifest.json`.

ENG1. Legacy Godot 4.7.1 replays authored rejection/commit against prior/offline hashes; captures show board/hints/coach/chain prompt and dashboard checks. Core 52/52 plus four extensions establish engine-local conformance, not transport, participants, narrative, usability, or efficacy.

F. Single-task Codex development observation

Frozen `ontology-map-20261023-01` covers one read-only bilingual dashboard task (n=1, independent repeats

TABLE XI
RAW-TRACE OBSERVATIONS FOR THE THREE DISCLOSURE ARMS WITH SEPARATE NORMAL, BOUNDARY, AND MALFORMED DENOMINATORS.

Arm	n (N/B/M)	Allow/Deny	Semantic mismatch	State/Display fail	Replay fail	Unique unsafe
unchecked_isolated	48/4/8	60/0	60	0/0	0	51
general_rule	48/4/8	16/44	0	0/0	0	0
ontology_gate	48/4/8	16/44	0	0/0	0	0

N/B/M = normal/boundary/malformed. Semantic mismatch compares the complete expected outcome; unique unsafe cases are the union of unauthorized allows and redisplay mutations. Permitted fresh-disclosure representation mismatches alone are not unsafe. Unchecked observations do not establish efficacy, superiority, or generalization.

TABLE XII
RULE-CHANGE FIXTURE TRACEABILITY AND EDIT SCOPE FOR PAIRED CASES; THE SHIPPED SCENARIO IS NOT MODIFIED.

Fixture	Mutation	Paired cases	Changed paths	Missed cases	Match
RC-001-stage-gate-2-to-1	disclosure_policy.stage_gates[0].minimum_stage 2→1	N-043, N-044	disclosure_policy.stage_gates[0].minimum_stage	none	PASS
RC-002-forbid-tide-marks-hint	disclosure_policy.permanently_forbidden ['keeper_betrayal']→['keeper_betrayal', 'tide_marks_hint']	N-047, N-048	disclosure_policy.permanently_forbidden[1]	none	PASS

Each fixture applies the same mutation to general_rule and ontology_gate. PASS requires no missed case and mutation_scope_preserved=true.

0), not an NPC trial/cohort. Six native CLI invocations (initial + five resumes; one read-only/no-edit) form one case. `turn_context.model=gpt-6.1-sol` is runtime selection, not provider attestation; provider identity/revision/seed/cost are `null`, and model-only latency is unavailable. Wall-clock 5942.665 s and CLI 263.32 s are neither inference latency nor the future endpoint.

Invocation six fixed inherited 660 px mobile-table minimum width: geometry RED (2 failed/35 passed), followed by 62/62 map/signal/Web checks (including cell/paragraph geometry) and a successful Godot Web build. Browser attempt 10 passed; 01–08 failed setup/routes; 09 passed older gates with clipping. Final $(\text{seq}, t_{\text{ms}})$: stage-0 denial $(68, 19468)$ (equal hashes), fresh disclosure $(239, 58234)$ (different), authorized redisplay $(249, 59793)$ (equal). At 1440×900/390×900, Enter, inspection/frozen-comparison invariance, and zero pre-navigation console/page/network errors/warnings were observed. Dated QA10 PASS differs from additive overall **NOT_PASS**: seven component gates passed; pointer-lock/yaw-guard and four pointer-lock page errors remained. This bounded integration is neither full-game creation, Python–Godot authorization round trip, nor a model outcome. Privacy/source-binding failures and same-thread repairs add no n; Jev is `proposal_only`, without label/approval authority.

G. Fixed-game native runtime comparison

This public packet retains 8 assigned rows for one authored quest (`sealed-lighthouse-v1`), with seed 20261024 and two repeats per selector/representation cell. System-endpoint successes are 5/8; observed native executions are 7. Host invocations are summed from actual receipts (23), and observed repairs sum to 0. Committed transitions total 16, of which unsafe are 0; semantic replay valid/invalid/NR is 7/0/1. Fresh-engine reexecution disposition counts are VALID=6; INVALID=0; NOT_RUN=1; NOT_OBSERVED=1. Observed elapsed time is n=7; 48.19/41.09 seconds (mean/median), and preparation time is n=7; 6.67/6.30 seconds. Unreported values are NR, not zero.

The first row is retained as a harness JSON-serialization technical failure and is not attributed to GPT-6.1 capability. At least 2 GPT-6.1 ontology row(s) retain a response-timeout failure under the frozen 60-second bound. Models provide typed-intent proposals; the deterministic Python planner/checker and Godot commit retain authority. The table is therefore a technical observation for one quest, not an efficacy, superiority, noninferiority, general ontology-consistency, or player-benefit claim. Semantic record-link replay is not fresh-engine reexecution. Provider identity/revision/cost and provider-request counts remain NR outside the public observation boundary.

VII. DISCUSSION, THREATS, ETHICS, AND ARTIFACT SCOPE

Semantic repair [48], [55], [56], [57] uses pre-execution errors; the oracle only reads state. Rule agreement/edit-scope diagnostics are neither capability nor ontology superiority. Implementation confound: row 1 failed under controller-v3 before a model response; rows 2–8 used controller-v4 after a one-line JSON-boundary correction with unchanged model-visible inputs, so selector/representation differences are not model-attributable.

The fixed-game pilot is bounded native evidence; no ontology-efficacy test ran. A (model/stack consistency) and C (fresh-reference noninferiority) remain unexecuted and require budget-bounded completion under fixed source/policy, sealed assignments, exact identity/revision fields, and separate safety/replay accounting; it answers neither. Endpoint time is request/evidence receipt to the first receipt passing frozen mandatory checks (time-to-first-all-predeclared-validation-pass); unfinished/timeout/failure/stage-not-reached runs have no successful time.

Threats and boundaries. The native pilot has one authored quest and two repeats per cell; comparator power, neutral complete-attempt inventories, and independent traceability scores are absent. State consistency implies neither behavioral nor narrative coherence [58]. Predicate/role/Godot/DISC sets are not latent states; declarations are not human gold/holdouts;

TABLE XIII
FIXED-GAME NATIVE RUNTIME COMPARISON (N=2 PER CELL). ALL EIGHT ASSIGNMENTS, INCLUDING THE HARNESS FAILURE AND TIMEOUTS, REMAIN IN DENOMINATORS.

Requested selector	Form	Goal/assigned	Observed exec.	Host calls	Unsafe/committed	Elapsed median (n)	Fresh engine
gpt-6.1-sol	general_rule	1/2	1	4	0/3	45.91 (1)	1/0/0/1
gpt-6.1-sol	ontology	0/2	2	3	0/1	72.35 (2)	1/0/1/0
gpt-5.6-sol	general_rule	2/2	2	8	0/6	40.24 (2)	2/0/0/0
gpt-5.6-sol	ontology	2/2	2	8	0/6	33.11 (2)	2/0/0/0

Fresh engine counts follow VALID/INVALID/NOT_RUN/NOT_OBSERVED order. NR is unreported, not zero. Overall counts and the separate semantic replay are reported in the text.

lanes retain separate denominators. No participant, affect, comprehension, agency, enjoyment, immersion, benefit, population, safety-rate, model-ranking, or commercial-performance inference follows. Hashes do not authenticate; replay equality is not semantic correctness. Human/affect studies require ethics review, consent, compensation disclosure, minimization, retention/deletion, opt-out, and calibrated judges [40], [41], [42], [43].

VIII. CONCLUSION

TRACE-RPG combines bounded C1–C5 authorization/repair/replay/accounting with a source-bound bilingual observer. The Codex case/native pilot retain failures and boundaries, not model-performance evidence. Whole-game, timing, traceability, confirmatory A/C, and repeated-cross-model evaluation remain future; historical cohorts stay unpooled.

DISCLOSURE OF AI USAGE

LLMs assisted prose/code/tests/citations; hosted candidates were labelled. Curated AI art/desk mesh has file provenance; snapshots/hashes/ledger are presentation-only. Authors checked sources and accept responsibility; NPC inputs/tables retain assignment/protocol/schema hashes and unknown token/cost/model-revision fields.

DATA AND CODE AVAILABILITY

Implementation/build source, frozen packets, Godot slice, deterministic evaluation, and machine-checked contribution, reference-topic, and experiment matrices: research/academic-pipeline/. DEV-CASE: research/development-cases/ontology-map-20261023-01/. Authoritative public native bundle (`results.json`, `public-traces.json`, `integrity-audit.json`): research/runtime-model-swap-20261024/ codex-live-20261026/public-results-v1/; generated runtime tables/results and trace-derived figure: paper/latex/generated/, paper/latex/figures/. Proposed rubric/codebook/source manifest and future whole-game/repeated-cross-model plan: research/traceability-scoring-20261023/, research/development-comparison-20261023/. Frozen disclosure inputs and standalone signal/NPC artifacts: paper/latex/generated/, paper/latex/figures/; earlier directories are historical receipts, not current-runtime proof; no reviewer archive/DOI deposit claimed. Proposed, unmeasured Table XIV is in traceability-supplement.pdf (matching .tex source) under paper/latex/en/ and paper/latex/ko/.

REFERENCES

[1] M.-A. Côté, Á. Kádár, X. Yuan *et al.*, "TextWorld: A learning environment for text-based games," in *Computer Games*, ser. Communications in Computer and Information Science. Springer International Publishing, 2019, vol. 1017, pp. 41–75.

[2] R. Wang, P. Jansen, M.-A. Côté, and P. Ammanabrolu, "ScienceWorld: Is your agent smarter than a 5th grader?" in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2022, pp. 11 279–11 298.

[3] N. Weir, R. Thomas, R. d'Amore, K. Hill, B. Van Durme, and H. Jhamtani, "Ontologically faithful generation of non-player character dialogues," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2024, pp. 9212–9242.

[4] T. Ashby, B. K. Webb, G. Knapp, J. Searle, and N. Fulda, "Personalized quest and dialogue generation in role-playing games: A knowledge graph- and language model-based approach," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, 2023, pp. 1–20.

[5] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. ACM, 2023, pp. 1–22.

[6] Y. Shao, L. Li, J. Dai, and X. Qiu, "Character-LLM: A trainable agent for role-playing," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2023, pp. 13 153–13 187.

[7] N. Akoury, Q. Yang, and M. Iyyer, "A framework for exploring player perceptions of LLM-generated dialogue in commercial video games," in *Findings of the Association for Computational Linguistics: EMNLP 2023*. Association for Computational Linguistics, 2023, pp. 2295–2311.

[8] M. Hochreiter, S. Kriglstein, and G. Wallner, "Beyond pre-defined scripts: Player perceptions on generative non-player character dialogues," in *Proceedings of the 31st International Conference on Intelligent User Interfaces*. ACM, 2026, pp. 2004–2018.

[9] M. Yin, H. Zu, W. Gu *et al.*, "How contextualized generative AI shapes player experience in games," *Entertainment Computing*, vol. 58, p. 101194, 2026.

[10] V. Figueiredo and D. Elumeze, "Symbolically scaffolded play: Designing role-sensitive prompts for generative NPC dialogue," 2025, preprint.

[11] G. N. Yannakakis and J. Togelius, "Experience-driven procedural content generation," *IEEE Transactions on Affective Computing*, vol. 2, no. 3, pp. 147–161, 2011.

[12] D. Melhart, D. Gravina, and G. N. Yannakakis, "Moment-to-moment engagement prediction through the eyes of the observer: PUBG streaming on Twitch," in *FDG '20: Proceedings of the 15th International Conference on the Foundations of Digital Games*. ACM, 2020, pp. 1–10.

[13] Z. Wang, Q. Guo, R. Singh, and X. Hu, "Do vision language models understand human engagement in games?" 2026, preprint.

[14] L. Fan, G. Wang, Y. Jiang *et al.*, "MineDojo: Building open-ended embodied agents with internet-scale knowledge," in *Advances in Neural Information Processing Systems*, vol. 35. Neural Information Processing Systems Foundation, 2022, pp. 18 343–18 362.

[15] D. Pérez-Liébana, J. Liu, A. Khalifa, R. D. Gaina, J. Togelius, and S. M. Lucas, "General video game AI: A multitrack framework for evaluating agents, games, and content generation algorithms," *IEEE Transactions on Games*, vol. 11, no. 3, pp. 195–214, 2019.

[16] X. Liu, H. Yu, H. Zhang *et al.*, "AgentBench: Evaluating LLMs as agents," in *International Conference on Learning Representations*, 2024, pp. 52 989–53 046.

[17] C. Ma, J. Zhang, Z. Zhu *et al.*, "AgentBoard: An analytical evaluation board of multi-turn LLM agents," in *Advances in Neural Information Processing Systems*, vol. 37. Neural Information Processing Systems Foundation, 2024, pp. 74 325–74 362.

[18] X. Zhou, H. Zhu, L. Mathur *et al.*, "SOTOPIA: Interactive evaluation for social intelligence in language agents," in *International Conference on Learning Representations*, 2024, pp. 40 975–41 019.

[19] D. Paglieri, B. Cupiał, S. Coward *et al.*, "BALROG: Benchmarking agentic LLM and VLM reasoning on games," in *International Conference on Learning Representations*, 2025, pp. 96 666–96 702.

[20] P. Le Pelletier de Woillemont, R. Labory, and V. Corruble, "Automated play-testing through RL based human-like play-styles generation," *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 18, no. 1, pp. 146–154, 2022.

[21] Y. Ma, J. Yan, B. Shi *et al.*, "Can LLM agents stick to the script? a benchmark for long-horizon consistency in interactive narratives," 2026, preprint (v1); ICML 2026 acceptance comment on arXiv, proceedings record unverified.

[22] S. Geng, M. Josifoski, M. Peyrard, and R. West, "Grammar-constrained decoding for structured NLP tasks without finetuning," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2023, pp. 10 932–10 952.

[23] L. Beurer-Kellner, M. Fischer, and M. Vechev, "Prompting is programming: A query language for large language models," *Proceedings of the ACM on Programming Languages*, vol. 7, no. PLDI, pp. 1946–1969, 2023.

[24] T. Scholak, N. Schucher, and D. Bahdanau, "PICARD: Parsing incrementally for constrained auto-regressive decoding from language models," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2021, pp. 9895–9901.

[25] S. Kambhampati, K. Valmeekam, L. Guan *et al.*, "Position: LLMs can't plan, but can help planning in LLM-Modulo frameworks," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 22 895–22 907.

[26] B. Ichter, A. Brohan, Y. Chebotar *et al.*, "Do as I can, not as I say: Grounding language in robotic affordances," in *Proceedings of the 6th Conference on Robot Learning (CoRL 2022)*, ser. Proceedings of Machine Learning Research, vol. 205. PMLR, 2023, pp. 287–318.

[27] H. Wang, C. M. Poskitt, and J. Sun, "AgentSpec: Customizable runtime enforcement for safe and reliable LLM agents," 2025, preprint (v3); ICSE 2026 acceptance stated on arXiv, proceedings record unverified.

[28] Z. Zeng, S. Li, J. Xi, A. Zhu, and P. Ammanabrolu, "Setting the DC: Tool-grounded D&D simulations to test LLM agents," in *NeurIPS 2025 Workshop on Generative and Protective AI for Content Creation (GenProCC)*, 2025, non-archival workshop paper; OpenReview/PDF list Zeng first, with Zeng and Li as equal contributors; NeurIPS/GenProCC lists display Li first.

[29] E. Zhou, S. Basavatia, M. Siam, Z. Chen, and M. O. Riedl, "STORY2GAME: Generating (almost) everything in an interactive fiction game," 2025, preprint.

[30] S. Buongiorno, L. Klinkert, Z. Zhuang, T. Chawla, and C. Clark, "PANGeA: Procedural artificial narrative using generative AI for turn-based, role-playing video games," *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 20, no. 1, pp. 156–166, 2024.

[31] M. Vaucher, S. Silveira, S. Góngora, and L. Chiruzzo, "IVIE: A neuro-symbolic approach to incremental and validated generation of interactive fiction worlds," in *Proceedings of the 17th International Conference on Computational Creativity (ICCC'26)*. Association for Computational Creativity, 2026, archival proceedings (ISBN); preprint arXiv:2606.13348.

[32] S. Góngora, L. Chiruzzo, G. Méndez, and P. Gervás, "World-state transformations for neuro-symbolic interactive storytelling," 2026, preprint.

[33] S. Geng, H. Cooper, M. Moskal *et al.*, "JSONSchemaBench: A rigorous benchmark of structured outputs for language models," 2025, preprint (v3).

[34] X. He, Y. Tian, Y. Sun *et al.*, "G-Retriever: Retrieval-augmented generation for textual graph understanding and question answering," in *Advances in Neural Information Processing Systems*, vol. 37. Neural Information Processing Systems Foundation, 2024, pp. 132 876–132 907.

[35] B. Jiménez Gutiérrez, Y. Shu, Y. Gu, M. Yasunaga, and Y. Su, "HippoRAG: Neurobiologically inspired long-term memory for large language models," in *Advances in Neural Information Processing Systems*, vol. 37. Neural Information Processing Systems Foundation, 2024, pp. 59 532–59 569.

[36] P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav, "Mem0: Building production-ready AI agents with scalable long-term memory," in *ECAI 2025*, ser. Frontiers in Artificial Intelligence and Applications, vol. 413. IOS Press, 2025, pp. 2993–3000.

EXTERNAL TRACE COUNTS

	N normal	B boundary	M malformed
U	E 0/48 n=48 R 48/0 C 0/0 NR 48	E 0/4 n=4 R 4/0 C 0/0 NR 4	E 0/8 n=8 R 8/0 C 0/0 NR 8
G	E 48/0 n=48 R 48/0 C 14/68 NR 0	E 4/0 n=4 R 4/0 C 2/4 NR 0	E 8/0 n=8 R 8/0 C 0/16 NR 0
O	E 48/0 n=48 R 48/0 C 14/68 NR 0	E 4/0 n=4 R 4/0 C 2/4 NR 0	E 8/0 n=8 R 8/0 C 0/16 NR 0

U unchecked_isolated · G general_rule · O ontology_gate

E expected P/F (n cases) · R replay P/F

C records P/F · NR cases with no reported checks

P pass · F fail; NR unreported, not pass/skipped/out-of-scope.

Fig. 4. NPC-DISC-1 counts by arm/stratum. E/R: expected-match/replay; C: policy-check PASS/FAIL; NR: no reported check, not skipped. Packet-bound engineering counts measure neither latent state, free-language leakage, ontology superiority, nor player benefit.

FIXED-GAME NATIVE RUNTIME COMPARISON I 8 ASSIGNED ROWS

One authored quest; n=2 per selector/representation cell; NR = not reported, not zero. Labels remain explicit; color is supplementary.

Row	Requested selector	Representation / repeat	Status	Goal	Host calls	Elapsed (s)	Safety R record-link / E engine
1	gpt-6.1-sol	general_rule / r1	HARNESS	NR	0	NR	NOT_OBSERVED R NR / E NOT_OBSERVED
2	gpt-6.1-sol	general_rule / r2	SUCCESS	YES	4	45.91	PASS R YES / E VALID
3	gpt-6.1-sol	ontology / r1	TIMEOUT	NO	2	84.66	PASS R YES / E VALID
4	gpt-6.1-sol	ontology / r2	TIMEOUT	NR	1	60.04	NO_COMMIT_OBSERVED R YES / E NOT_RUN
5	gpt-5.6-sol	general_rule / r1	SUCCESS	YES	4	41.09	PASS R YES / E VALID
6	gpt-5.6-sol	general_rule / r2	SUCCESS	YES	4	39.39	PASS R YES / E VALID
7	gpt-5.6-sol	ontology / r1	SUCCESS	YES	4	36.68	PASS R YES / E VALID
8	gpt-5.6-sol	ontology / r2	SUCCESS	YES	4	29.55	PASS R YES / E VALID

Endpoint outcomes, receipt-derived host calls, and dispatch-to-terminal elapsed time.

Typed-intent proposal; deterministic planner/checker and Godot commit retain authority.

No efficacy/superiority/noninferiority/player-benefit claim; provider metadata and request counts remain NR.

Fig. 5. Trace-derived native runtime comparison for the fixed-game pilot (one authored quest; 8 assigned rows, 7 observed executions). System-goal outcomes and assigned denominators retain unknown provider identity, revision, cost, and request-count fields; one pre-response harness-serializer technical failure and two frozen 60-s `gpt-6.1-sol` ontology response-timeout failures are distinguished from goal successes. Evidence image, not gameplay screenshot.

[37] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," in *Advances in Neural Information Processing Systems*, vol. 34. Curran Associates, Inc., 2021, pp. 29 304–29 320.

[38] J. Pineau, P. Vincent-Lamarre, K. Sinha *et al.*, "Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program," *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021.

[39] P. Henderson, J. Hu, J. Romoff, E. Brunskill, D. Jurafsky, and J. Pineau, "Towards the systematic reporting of the energy and carbon footprints of machine learning," *Journal of Machine Learning Research*, vol. 21, no. 248, pp. 1–43, 2020.

[40] C. van der Lee, A. Gatt, E. van Miltenburg, S. Wubben, and E. Krahmer, "Best practices for the human evaluation of automatically generated text," in *Proceedings of the 12th International Conference on Natural Language Generation*. Association for Computational Linguistics, 2019, pp. 355–368.

[41] M. Karpinska, N. Akoury, and M. Iyyer, "The perils of using Mechanical Turk to evaluate open-ended text generation," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2021, pp. 1265–1285.

[42] L. Zheng, W.-L. Chiang, Y. Sheng *et al.*, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in *Advances in Neural Information Processing Systems*, vol. 36. Neural Information Processing Systems Foundation, 2023, pp. 46 595–46 623.

[43] G. H. Chen, S. Chen, Z. Liu, F. Jiang, and B. Wang, "Humans or LLMs as the judge? a study on judgement bias," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2024, pp. 8301–8327.

[44] D. Lakens, A. M. Scheel, and P. M. Isager, "Equivalence testing for psychological research: A tutorial," *Advances in Methods and Practices in Psychological Science*, vol. 1, no. 2, pp. 259–269, 2018.

[45] D. J. Barr, R. Levy, C. Scheepers, and H. J. Tily, "Random effects structure for confirmatory hypothesis testing: Keep it maximal," *Journal of Memory and Language*, vol. 68, no. 3, pp. 255–278, 2013.

[46] E. Feng, W. Zhou, Z. Liu *et al.*, "Get experience from practice: LLM agents with record & replay," 2025, preprint.

[47] Y. Sun, H. Wang, D. Li, G. Wang, and H. Zhang, "The emperor's new clothes in benchmarking? a rigorous examination of mitigation strategies for LLM benchmark data contamination," 2025, preprint (v1).

[48] A. Madaan, N. Tandon, P. Gupta *et al.*, "Self-Refine: Iterative refinement with self-feedback," in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023, pp. 46 534–46 594.

[49] M. O. Riedl, C. J. Saretto, and R. M. Young, "Managing interaction between users and agents in a multi-agent storytelling environment," in *Proceedings of the Second International Joint Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2003, pp. 741–748.

[50] R. E. Fikes and N. J. Nilsson, "STRIPS: A new approach to the application of theorem proving to problem solving," *Artificial Intelligence*, vol. 2, no. 3–4, pp. 189–208, 1971.

[51] F. B. Schneider, "Enforceable security policies," *ACM Transactions on Information and System Security*, vol. 3, no. 1, pp. 30–50, 2000.

[52] T. Haerder and A. Reuter, "Principles of transaction-oriented database recovery," *ACM Computing Surveys*, vol. 15, no. 4, pp. 287–317, 1983.

[53] R. Evans and E. Short, "Versu—a simulationist storytelling system," *IEEE Transactions on Computational Intelligence and AI in Games*, vol. 6, no. 2, pp. 113–130, 2014, predecessor title of IEEE Transactions on Games.

[54] S. G. Ware and C. Siler, "Sabre: A narrative planner supporting intention and deep theory of mind," in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, vol. 17, no. 1, 2021, pp. 99–106.

[55] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, "Reflexion: Language agents with verbal reinforcement learning," in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023, pp. 8634–8652.

[56] X. Chen, M. Lin, N. Schärli, and D. Zhou, "Teaching large language models to self-debug," in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024, accepted poster; OpenReview archival record; preprint arXiv:2304.05128.

[57] A. Solar-Lezama, L. Tancau, R. Bodík, S. Seshia, and V. Saraswat, "Combinatorial sketching for finite programs," in *Proceedings of the 12th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*. ACM, 2006, pp. 404–415.

[58] Y. Huang, G. Chen, J. Yao *et al.*, "Beyond state consistency: Behavior consistency in text-based world models," 2026, preprint.